

Firewalls: Packet, Stateful, NGFW

The generations

Age	Type	Decision basis
1	Packet filter (stateless)	Header-only rules, per-packet
2	Stateful	Full connection state table (established/related)
3	Application (NGFW)	App inspection, identity, SSL/TLS visibility

Stateless is fine for tiny networks; stateful is the baseline; NGFW earns its price on app-layer needs.

Rule-style essentials

```
# Cisco ASA-ish / generic: order matters (first match wins)
access-list OUT permit tcp host 10.0.0.5 any eq 443
access-list OUT deny ip any any log
# pfSense-style:
WAN: allow 443, blocking everything else, egress to specific
LAN->WAN: allow; LAN->DMZ: only web-ports; DMZ->LAN: deny
```

Craft principles

Explicit deny by default (default deny, allow what's needed)

Least privilege (the smallest port/protocol/source set)

Order matters (specific rules before general)

Log the denials (a blocked thing tells you something's probing)

Egress filtering (most orgs forget outbound rules)

Review quarterly – the rule list rots silently

Typical segment topology

```
[Internet] ---- FW1 (perimeter) ---- DMZ: web, mail, vpn
              |----+- MPLS internal
              |   +- user VLAN
              +---- FW2 (internal segmented zones) --- DB, mgmt
```

What a firewall does NOT do

Not an anti-malware layer; not a behavioral IDS

Cannot stop malicious-but-port-443 apps (hence NGFW + IDS)

SSL/TLS breaking needed to 'see' encrypted malware – with cost/privacy tradeoffs

A firewall is the gate, not the guard room. The professional firewall designer thinks in *segments*, *least privilege*, and *logging* – not 'open port 80 so the web works'.

[← Module start](#)

[IDS/IPS: Suricata, Rules, Depl...](#)